



Planning and Configuring IP Addresses

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

On an IP network, to connect a PC to the Internet, you need to apply an IP address for the PC. An IP address is like a real-world address. It identifies a node on a network, and is used to find the destination. Global network communication is based on IP addresses.

An IP address is an attribute of an interface on a network device, not an attribute of the network device itself. To assign an IP address to a device is to assign an IP address to an interface of the device actually. If a device has multiple interfaces, generally each interface requires at least one IP address.

After completing this task, you will have a clear understanding of network planning and device IP address configuration, and be able to independently plan networks and configure device IP addresses.

1.2 Objectives

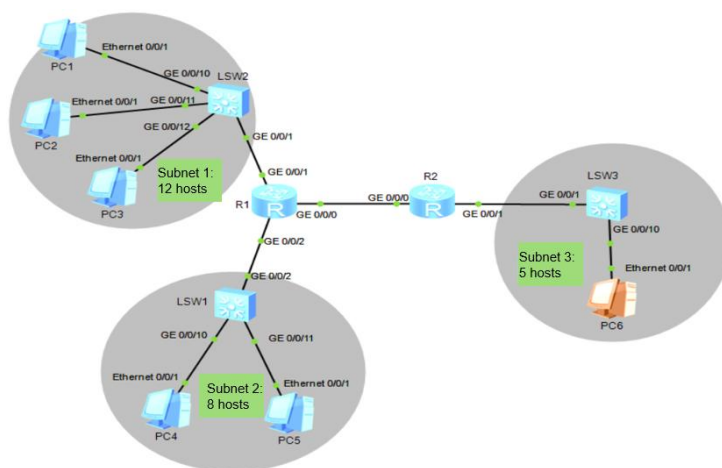
On completing this task, you will be able to:

1. Set up a network topology using Huawei eNSP.
2. Plan IP network segments.
3. Configure device IP addresses.
4. Perform tests.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



2.2 Initial Configuration

Huawei eNSP has been installed and is running properly.



3 Task Implementation

3.1 Setting Up a Topology

Step 1 Start eNSP.

After eNSP is started, the following window is displayed. Icons on the left pane indicate the products and devices supported by eNSP. The middle pane provides examples of various network scenarios. For details, see Figure 3-1.

Figure 3-1 eNSP startup window



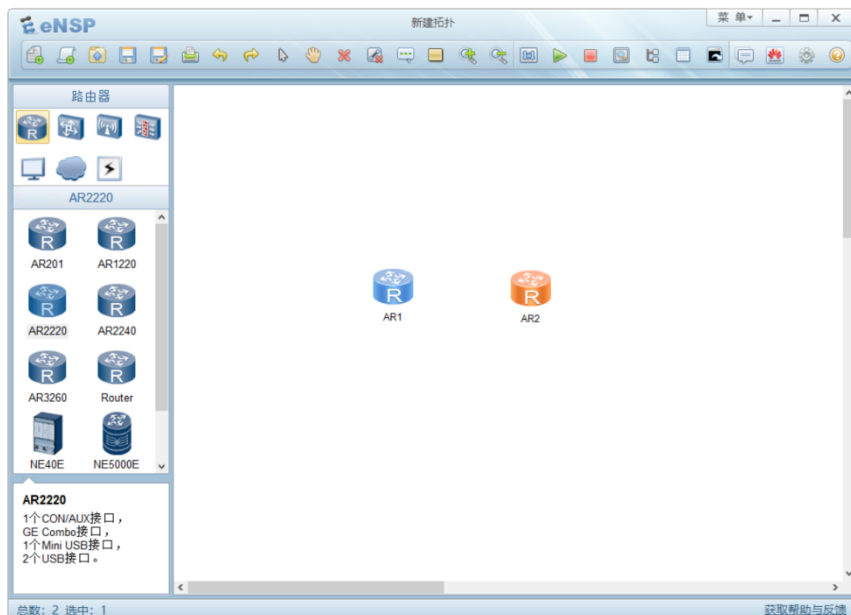
Click the create icon in the upper left corner of the window to create a lab scenario. On the blank area that is displayed, you can set up a network topology, practice networking, and analyze network behaviors. In this example, you need to set up a simple end-to-end network using two terminals.

Step 2 Set up a topology.

Click the router icon on the top of the left pane. Among the terminal devices displayed, select the **AR2220** icon and drag it to the blank area. See Figure 3-2.



Figure 3-2 Adding a router

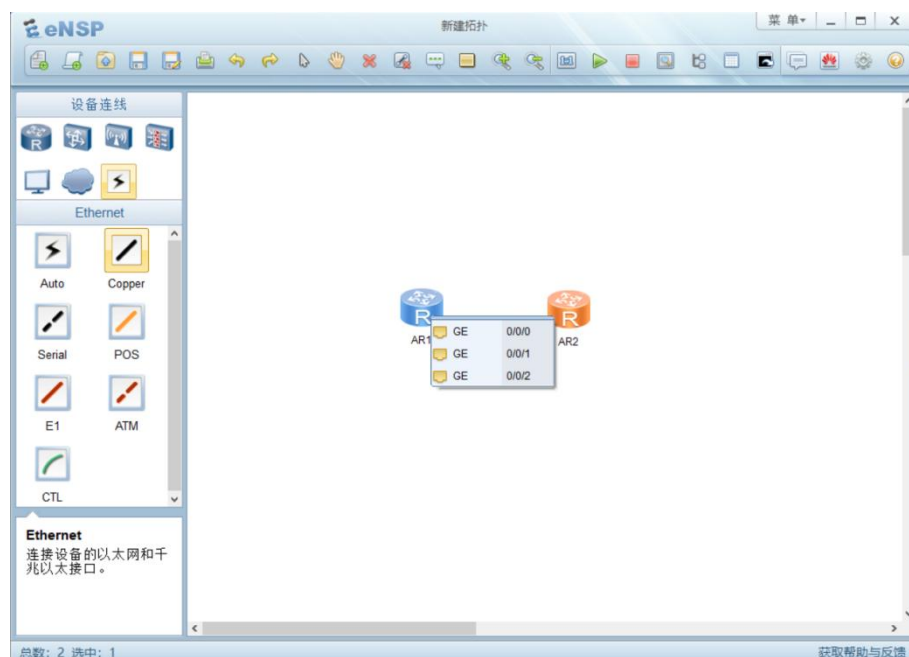


Repeat the preceding steps to drag another router icon to the blank area for setting up an end-to-end network topology.

Step 3 Establish a physical connection.

Click the device connection icon on the top of the left pane. Select the copper (Ethernet) icon among the media displayed. After you click the icon, the cursor represents a connector. Click the client device. All interfaces of the simulated device are displayed. Click **GE 0/0/0** to connect this interface. For details, see Figure 3-3.

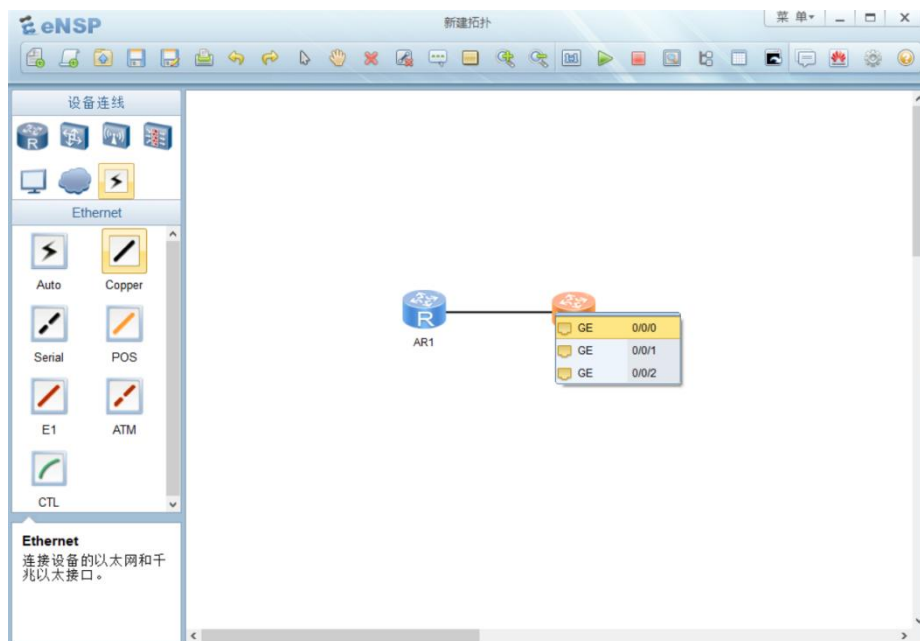
Figure 3-3 Establishing a physical connection





Click the other device and select the **GE 0/0/** interface as the endpoint of the connection. The physical connection between the two is then set up. See Figure 3-4.

Figure 3-4 Connecting the other router



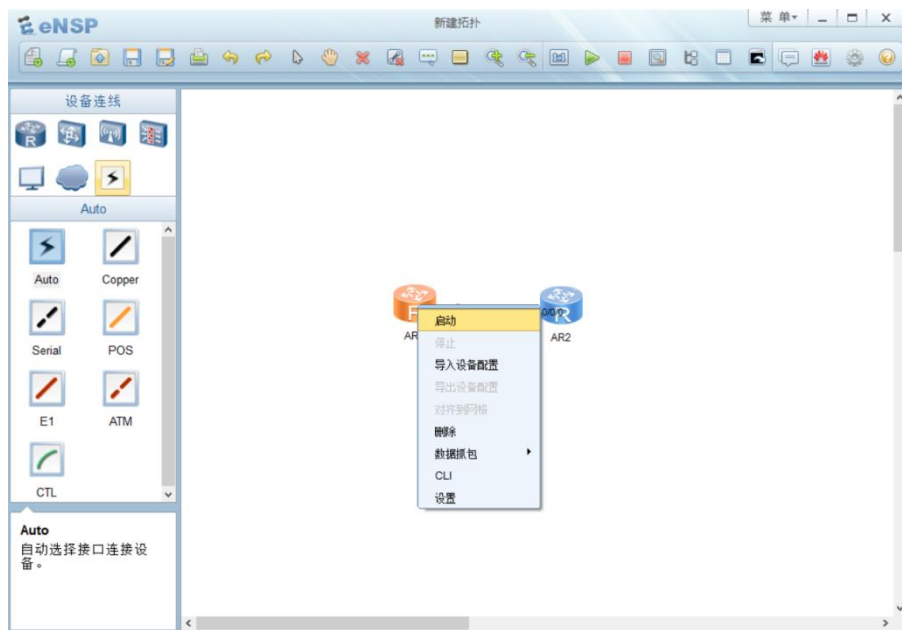
Two red dots are displayed at both ends of the connection on the established end-to-end network, indicating that the two connected interfaces are Down.

Step 4 Enter the router configuration page and start the router.

Right-click a terminal device. In the displayed attribute menu, select **Settings** to check the configuration of the device. Click **Start** to start the two routers. See Figure 3-5.



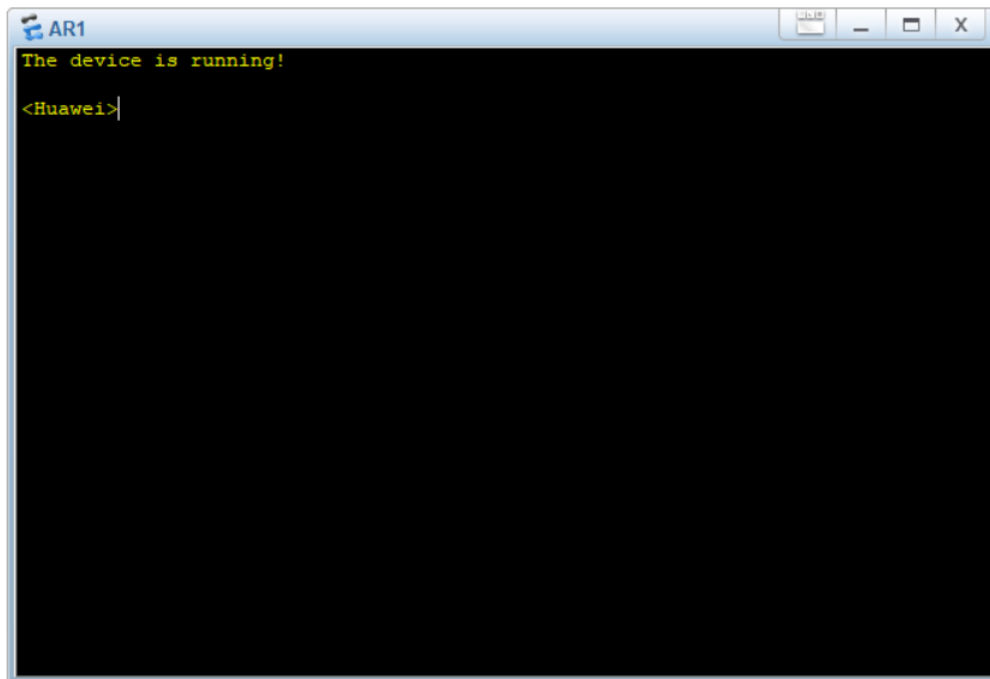
Figure 3-5 Starting the devices



Step 5 Enter the router configuration page.

Double-click the router icon. The router configuration page is displayed, as shown in Figure 3-6.

Figure 3-6 Router configuration page

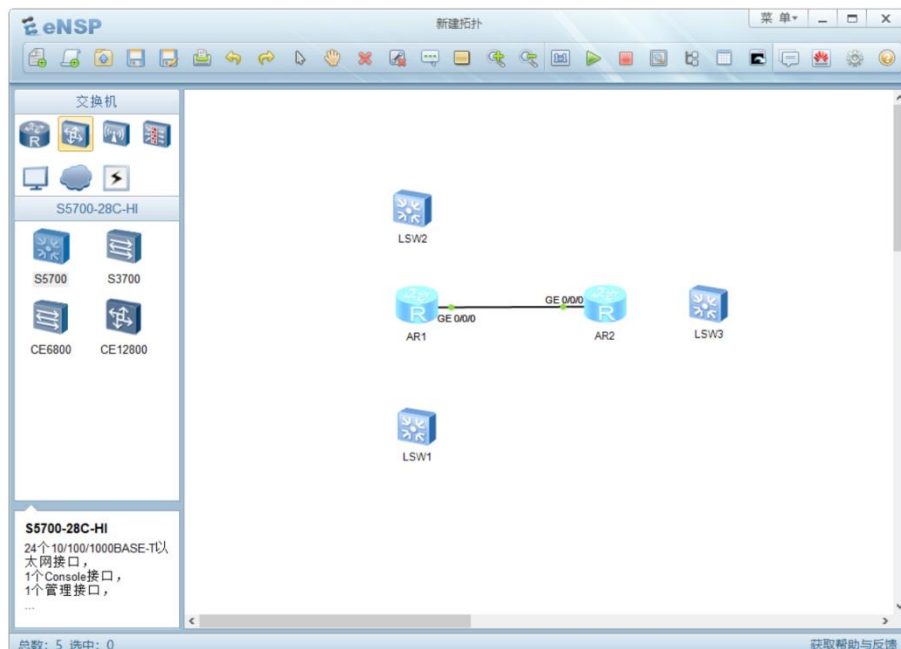




Step 6 Add switches.

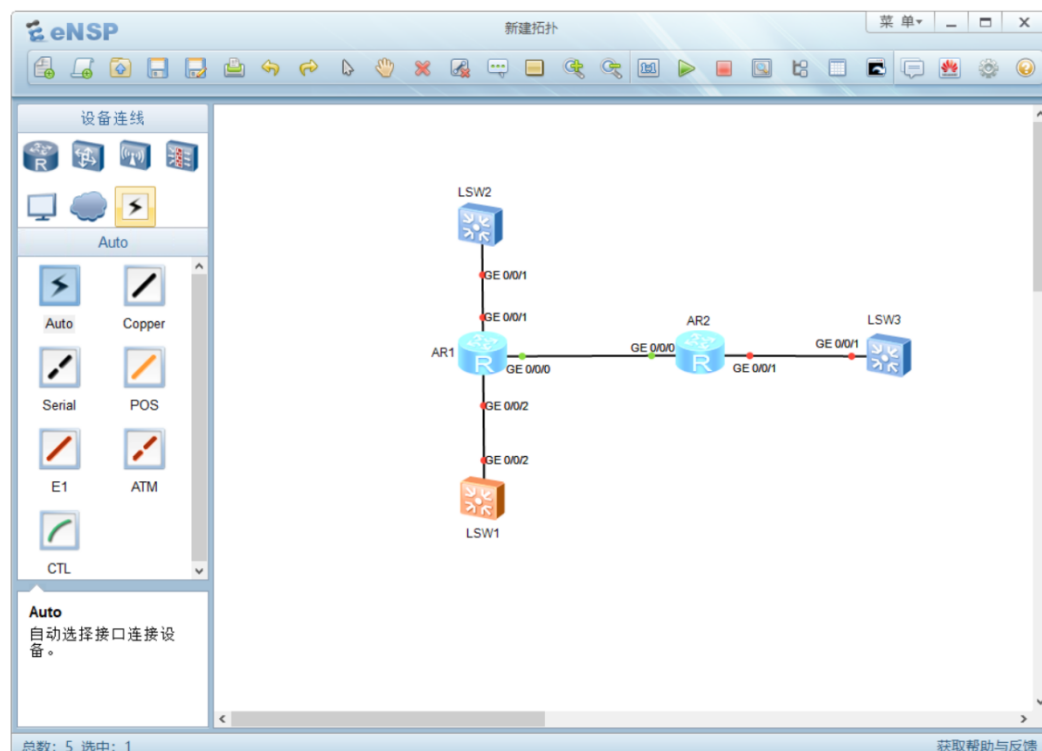
Add three switches in the same way, as shown in Figure 3-7.

Figure 3-7 Adding switches



Step 7 Connect the switches to the router. The connection interfaces are shown in Figure 3-8.

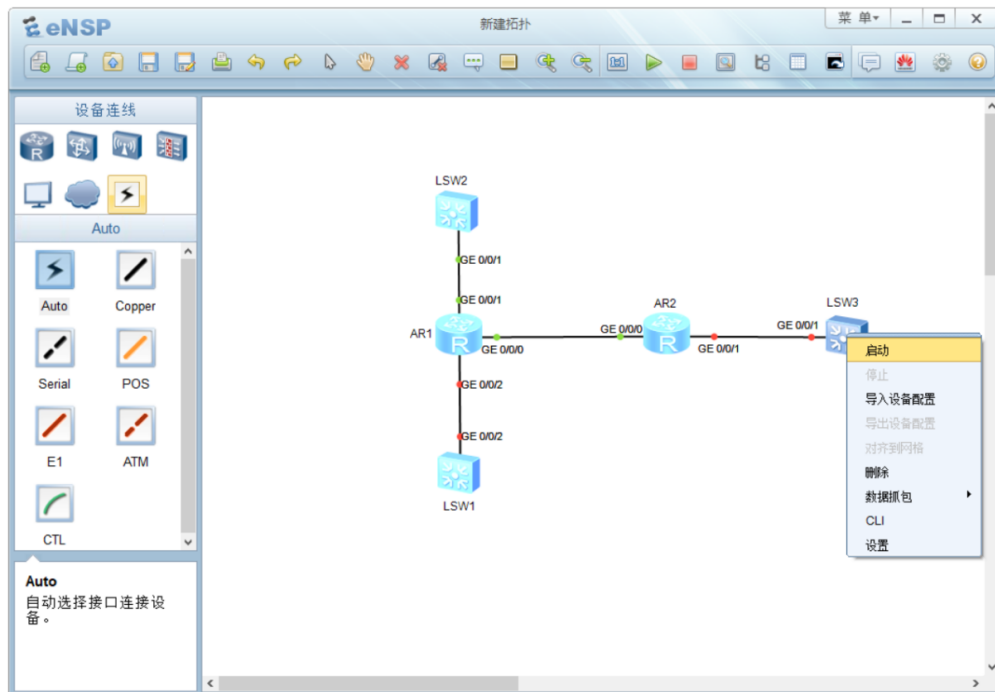
Figure 3-8 Connecting switches to the router





Step 8 Start the three switches, as shown in Figure 3-9.

Figure 3-9 Starting the switches

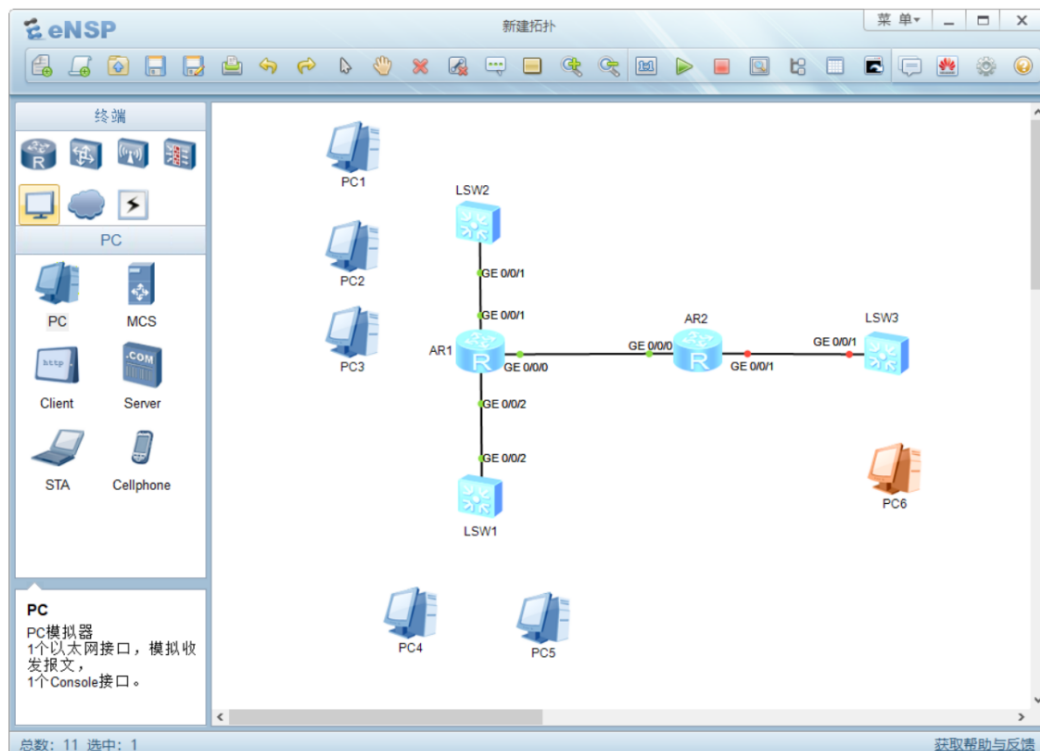


Step 9 Add PCs.

Add PCs for each switch to simulate different networks, as shown in Figure 3-10.

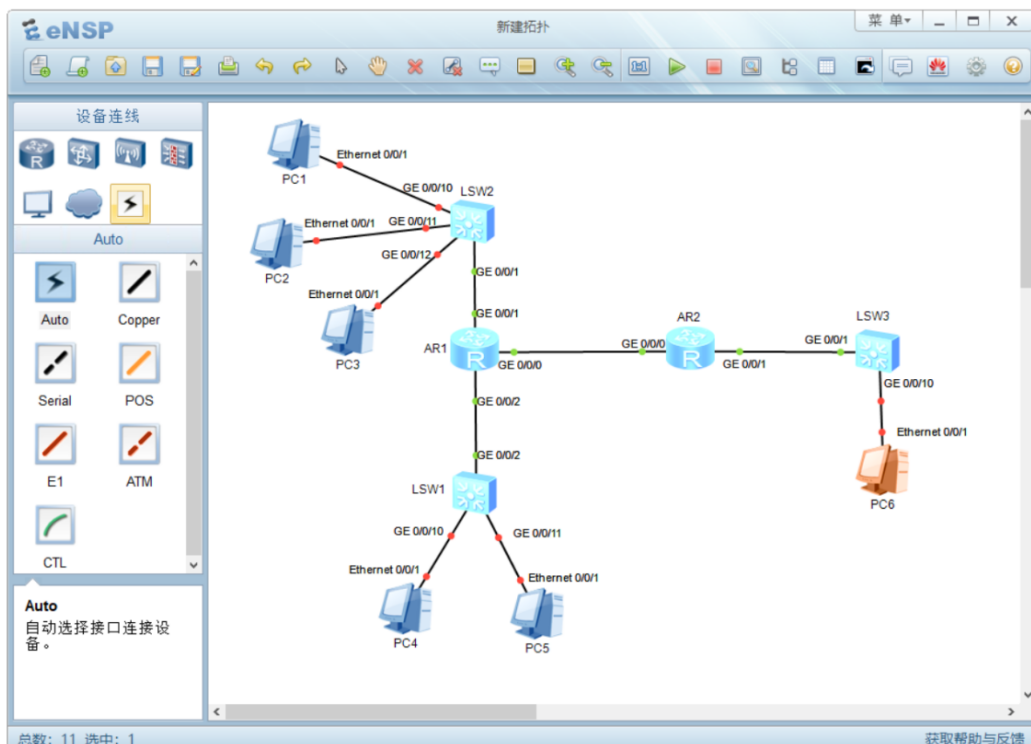


Figure 3-10 Adding PCs



Step 10 Connect PCs to the switches. The connection interfaces are shown in Figure 3-11.

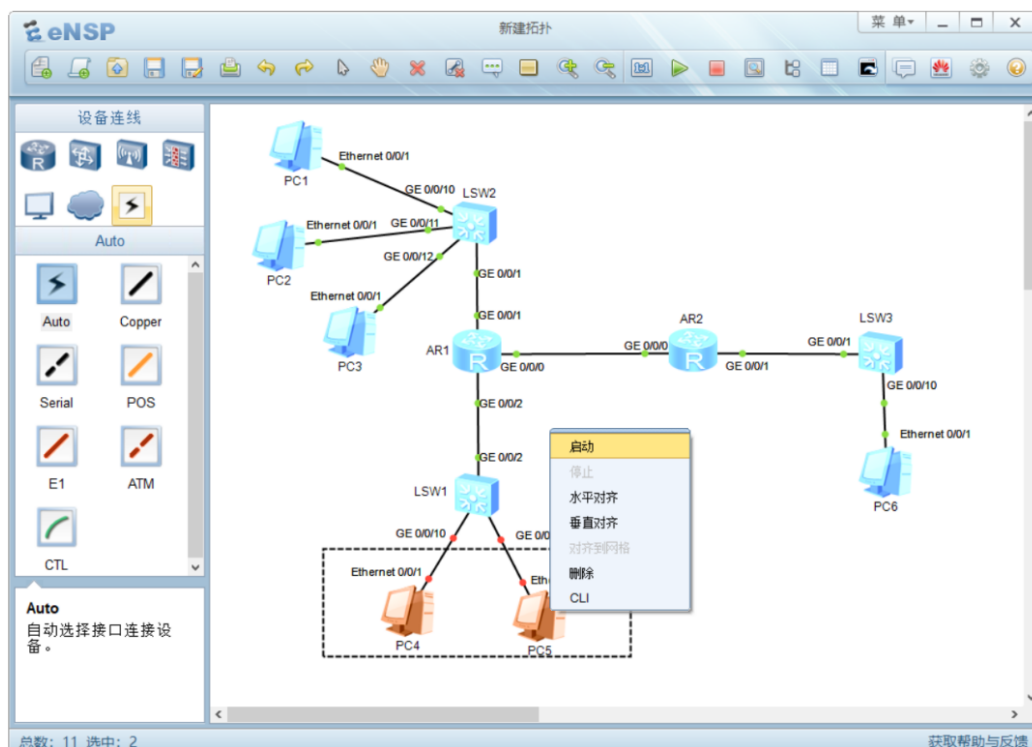
Figure 3-11 Connecting PCs to the switches





Step 11 Start all PCs, as shown in Figure 3-12.

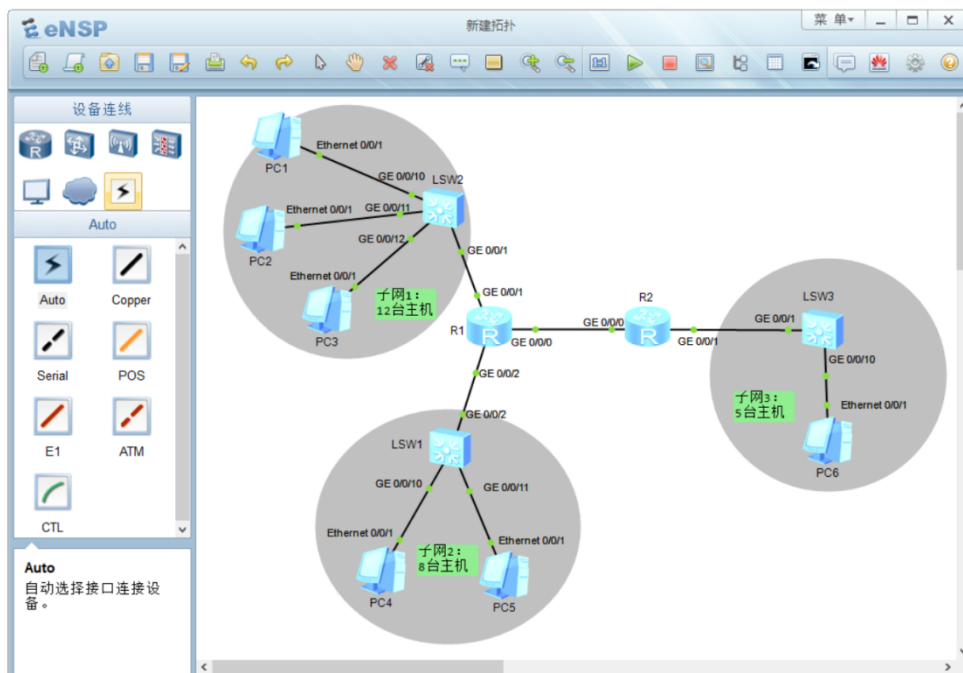
Figure 3-12 Starting PCs



----End

3.2 Planning IP Network Segments

Assume that there is a Class C network address segment 192.168.1.0/24. Use variable-length subnet masks to allocate IP addresses to three subnets, as shown in Figure 3-13.

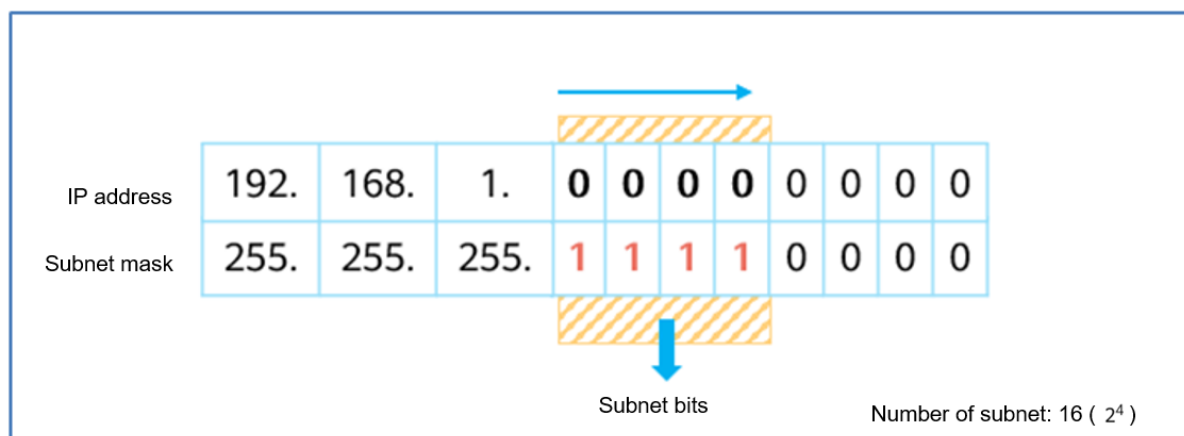
**Figure 3-13** Three subnets

Step 1 In practical network planning, prioritize subnets with more hosts for planning.

- Subnet 1 has 14 hosts and requires no less than 4 host bits ($2^n - 2 \geq 12$, with $n \geq 4$).
- Subnet 2 has 8 hosts and requires no less than 4 host bits ($2^n - 2 \geq 8$, with $n \geq 4$).
- Subnet 3 has 5 hosts and requires no less than 3 host bits ($2^n - 2 \geq 5$, with $n \geq 3$).

Step 2 Borrow bits from host bits.

In practical network planning, prioritize subnets with more hosts for planning. Plan subsets 1 and 2 first. See Figure 3-14.

Figure 3-14 Borrowing bits from host bits

The network addresses of the subnets are as follows:



192.168.1.0
192.168.1.16
192.168.1.32
192.168.1.48
192.168.1.64
192.168.1.80
192.168.1.96
192.168.1.112
192.168.1.128
192.168.1.144
192.168.1.160
192.168.1.176
192.168.1.192
192.168.1.208
192.168.1.224
192.168.1.240

Step 3 Select subnets. You can select two subnets from the preceding 16 subnets as required and configure them for subnet 1 and subnet 2. Here, 192.168.1.0/28 is selected for subnet 1, and 192.168.1.16/28 is for subnet 2.

Subnet 1: 192.168.1.0/28

Network	Mask	Host IP address	Gateway (router Interface)
192.168.1.0	255.255.255.240	192.168.1.2~15/28	GE 0/0/1 interface on R1 192.168.1.1/28

Subnet 2: 192.168.1.16/28

Network	Mask	Host IP address	Gateway (router Interface)
192.168.1.16	255.255.255.240	192.168.1.18~31/28	GE 0/0/2 interface on R1 192.168.1.17/28

Step 4 Select 192.168.1.32/28 for subnet 3. Subnet 3 requires only three host bits and one more host bit needs to be borrowed.

Subnet 3: 192.168.1.32/29

Network	Mask	Host IP address	Gateway (router Interface)
192.168.1.32	255.255.255.248	192.168.1.34~39/29	GE 0/0/1 interface on R2 192.168.1.33/29



Step 5 Configure the interconnection address of the two routers. Here, 10.1.12.0/24 is used.

Router R1	Interface on R1	Router R2	Interface on R2
10.1.12.1/24	GE 0/0/0	192.168.1.34~39/29	GE 0/0/0

Step 6 Configure the loopback interface on the two routers. A loopback interface is usually configured with a 32-bit mask.

Loopback interface of R1	Loopback interface of R2
1.1.1.1/32	2.2.2.2/32

----End

3.3 Configuring IP Addresses

Step 1 Configure interfaces on R1.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip add
[R1-GigabitEthernet0/0/1]ip address 192.168.1.1 28
[R1-GigabitEthernet0/0/1]quit
[R1]int g0/0/2
[R1-GigabitEthernet0/0/2]ip add
[R1-GigabitEthernet0/0/2]ip address 192.168.1.17 28
[R1-GigabitEthernet0/0/2]quit
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.1.12.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]
[R1]int LoopBack 0
[R1-LoopBack0]ip address 1.1.1.1 32
[R1-LoopBack0]quit
[R1]
```

Step 2 Configure interfaces on R2.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int LoopBack 0
[R2-LoopBack0]ip address 2.2.2.2 32
[R2-LoopBack0]quit
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 192.168.1.33 29
[R2-GigabitEthernet0/0/1]quit
```



```
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.1.12.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]
```

- Step 3** Configure IP addresses for the PCs on subnet 1. Select three IP addresses in the range of 192.168.1.2-15/28 and set the gateway address to 192.168.1.1. Figure 3-15 uses PC1 on subnet 1 as an example.

Figure 3-15 Configuring an IP address for a PC on subnet 1

The screenshot shows the 'PC1' configuration window with the following settings:

- 基础配置** (Basic Configuration) tab is selected.
- 主机名:** (Hostname) is empty.
- MAC 地址:** (MAC Address) is 54-89-98-0F-3E-F3.
- IPv4 配置** (IPv4 Configuration) section:
 - ☒ **静态** (Static) is selected.
 - ☐ DHCP is unselected.
 - ☐ 自动获取 DNS 服务器地址 (Automatically obtain DNS server address) is unselected.
 - IP 地址:** (IP Address) is 192 . 168 . 1 . 2.
 - 子网掩码:** (Subnet Mask) is 255 . 255 . 255 . 240.
 - 网关:** (Gateway) is 192 . 168 . 1 . 1.
 - DNS1:** is 0 . 0 . 0 . 0.
 - DNS2:** is 0 . 0 . 0 . 0.
- IPv6 配置** (IPv6 Configuration) section:
 - ☒ **静态** (Static) is selected.
 - ☐ DHCPv6 is unselected.
 - IPv6 地址:** (IPv6 Address) is ::.
 - 前缀长度:** (Prefix Length) is 128.
 - IPv6 网关:** (IPv6 Gateway) is ::.
- 应用** (Apply) button is at the bottom right.

- Step 4** Configure IP addresses for the PCs on subnet 2. Select three IP addresses in the range of 192.168.1.18-31/28 and set the gateway address to 192.168.1.17. Figure 3-16 uses PC4 on subnet 2 as an example.



Figure 3-16 Configuring an IP address for a PC on subnet 2

The screenshot shows the 'PC4' configuration window with the '基础配置' (Basic Configuration) tab selected. The '主机名' (Host Name) field is empty. The 'MAC 地址' (MAC Address) field is set to '54-89-98-08-0E-2B'. Under the 'IPv4 配置' (IPv4 Configuration) section, the '静态' (Static) radio button is selected. The 'IP 地址' (IP Address) is '192.168.1.18', '子网掩码' (Subnet Mask) is '255.255.255.240', and '网关' (Gateway) is '192.168.1.17'. The 'DNS1' and 'DNS2' fields are both set to '0.0.0.0'. The '自动获取 DNS 服务器地址' (Automatically obtain DNS server address) checkbox is unchecked. Under the 'IPv6 配置' (IPv6 Configuration) section, the '静态' (Static) radio button is selected. The 'IPv6 地址' (IPv6 Address) is '::', '前缀长度' (Prefix Length) is '128', and 'IPv6 网关' (IPv6 Gateway) is '::'. An '应用' (Apply) button is at the bottom right.

Step 5 Configure IP addresses for the PCs on subnet 3. Select three IP addresses in the range of 192.168.1.34-39/29 and set the gateway address to 192.168.1.33. Figure 3-17 uses PC6 on subnet 3 as an example.

Figure 3-17 Configuring an IP address for a PC on subnet 3

The screenshot shows the 'PC6' configuration window with the '基础配置' (Basic Configuration) tab selected. The '主机名' (Host Name) field is empty. The 'MAC 地址' (MAC Address) field is set to '54-89-98-07-16-2D'. Under the 'IPv4 配置' (IPv4 Configuration) section, the '静态' (Static) radio button is selected. The 'IP 地址' (IP Address) is '192.168.1.34', '子网掩码' (Subnet Mask) is '255.255.255.248', and '网关' (Gateway) is '192.168.1.33'. The 'DNS1' and 'DNS2' fields are both set to '0.0.0.0'. The '自动获取 DNS 服务器地址' (Automatically obtain DNS server address) checkbox is unchecked. Under the 'IPv6 配置' (IPv6 Configuration) section, the '静态' (Static) radio button is selected. The 'IPv6 地址' (IPv6 Address) is '::', '前缀长度' (Prefix Length) is '128', and 'IPv6 网关' (IPv6 Gateway) is '::'. An '应用' (Apply) button is at the bottom right.



4 Verification

1. On R1, ping the IP address 10.1.12.2 of the direct interface on R2.

```
<R1>ping 10.1.12.2
PING 10.1.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.1.12.2: bytes=56 Sequence=1 ttl=255 time=30 ms
  Reply from 10.1.12.2: bytes=56 Sequence=2 ttl=255 time=10 ms
  Reply from 10.1.12.2: bytes=56 Sequence=3 ttl=255 time=10 ms
  Reply from 10.1.12.2: bytes=56 Sequence=4 ttl=255 time=20 ms
  Reply from 10.1.12.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.1.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/18/30 ms
```

2. On R1, ping the IP address 192.168.1.2 of the PC on subnet 1.

```
<R1>ping 192.168.1.2
PING 192.168.1.2: 56 data bytes, press CTRL_C to break
  Reply from 192.168.1.2: bytes=56 Sequence=1 ttl=128 time=50 ms
  Reply from 192.168.1.2: bytes=56 Sequence=2 ttl=128 time=50 ms
  Reply from 192.168.1.2: bytes=56 Sequence=3 ttl=128 time=60 ms
  Reply from 192.168.1.2: bytes=56 Sequence=4 ttl=128 time=30 ms
  Reply from 192.168.1.2: bytes=56 Sequence=5 ttl=128 time=50 ms

--- 192.168.1.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/48/60 ms
```

3. On R1, ping the IP address 192.168.1.18 of the PC on subnet 2.

```
<R1>ping 192.168.1.18
PING 192.168.1.18: 56 data bytes, press CTRL_C to break
  Reply from 192.168.1.18: bytes=56 Sequence=1 ttl=128 time=40 ms
  Reply from 192.168.1.18: bytes=56 Sequence=2 ttl=128 time=40 ms
  Reply from 192.168.1.18: bytes=56 Sequence=3 ttl=128 time=50 ms
  Reply from 192.168.1.18: bytes=56 Sequence=4 ttl=128 time=60 ms
  Reply from 192.168.1.18: bytes=56 Sequence=5 ttl=128 time=60 ms

--- 192.168.1.18 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/50/60 ms

<R1>
```




4. On R2, ping the IP address 192.168.1.34 of the PC on subnet 3.

```
<R2>ping 192.168.1.34
PING 192.168.1.34: 56 data bytes, press CTRL_C to break
  Reply from 192.168.1.34: bytes=56 Sequence=1 ttl=128 time=40 ms
  Reply from 192.168.1.34: bytes=56 Sequence=2 ttl=128 time=60 ms
  Reply from 192.168.1.34: bytes=56 Sequence=3 ttl=128 time=40 ms
  Reply from 192.168.1.34: bytes=56 Sequence=4 ttl=128 time=40 ms
  Reply from 192.168.1.34: bytes=56 Sequence=5 ttl=128 time=50 ms

--- 192.168.1.34 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/46/60 ms

<R2>
```

5. Check interface statistics on R1.

```
<R1>display ip int
<R1>display ip interface g0/0/1
GigabitEthernet0/0/1 current state : UP
Line protocol current state : UP
The Maximum Transmit Unit : 1500 bytes
input packets : 7, bytes : 540, multicasts : 0
output packets : 7, bytes : 540, multicasts : 0
Directed-broadcast packets:
  received packets:          0, sent packets:          0
  forwarded packets:        0, dropped packets:        0
ARP packet input number:      1
  Request packet:            1
  Reply packet:              0
  Unknown packet:            0
Internet Address is 192.168.1.1/28
Broadcast address : 192.168.1.15
TTL being 1 packet number:    0
TTL invalid packet number:    0
ICMP packet input number:     7
  Echo reply:                5
  Unreachable:                0
  Source quench:              0
  Routing redirect:           0
  Echo request:                2
  Router advert:              0
  Router solicit:             0
  Time exceed:                 0
  IP header bad:               0
  Timestamp request:          0
  Timestamp reply:            0
  Information request:         0
  Information reply:           0
  Netmask request:            0
  Netmask reply:              0
```



Unknown type: 0

<R1>

5 Quiz

1. (Multiple-answer question) A company is assigned a Class C network address segment 192.168.20.0/24. Which of the following subnets can be allocated to a department that has 40 hosts? (AC)
 - A. 192.168.20.64/26
 - B. 192.168.20.64/27
 - C. 192.168.20.128/26
 - D. 192.168.20.190/26
2. What is the function of a subnet mask?

A 32-bit IP subnet mask is used to distinguish the network ID and host ID in an IP address. The network ID indicates a network or subnet, and the host ID indicates a host on the network or subnet.
3. What is the function of the TTL field in an IP packet header?

If a loop exists on a network, IP packets may circulate indefinitely and fail to reach their destinations. The TTL field limits the lifetime of IP packets, ensuring that packets that cannot reach their destinations are discarded.
4. What is the function of a gateway?

A gateway receives and processes packets sent by hosts on the local network segment and forwards the packets to the destination network segment.